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```
CREATE DATABASE workplace_db;
```

7.24

```
CREATE TABLE employee (empNo VARCHAR(10) NOT NULL PRIMARY KEY, fName VARCHAR(50) NOT NULL, lName VARCHAR(50) NOT NULL, address VARCHAR(100), DOB DATE, sex ENUM('M', 'F') DEFAULT 'M', position ENUM('Manager', 'Team Leader', 'Analyst', 'Software Developer') DEFAULT 'Manager', deptNo VARCHAR(10));
```

Query OK, 0 rows affected (0.15 sec)

```
DESCRIBE employee;
```

Field	Type	Null	Key	Default	Extra
empNo	varchar(10)	NO	PRI	NULL	
fName	varchar(50)	NO		NULL	
lName	varchar(50)	NO		NULL	
address	varchar(100)	YES		NULL	
DOB	date	YES		NULL	
sex	enum('M','F')	YES		M	
position	enum('Manager','Team Leader','Analyst','Software Developer')	YES		Manager	
deptNo	varchar(10)	YES		NULL	

8 rows in set (0.05 sec)

```
CREATE TABLE department (deptNo VARCHAR(10) NOT NULL PRIMARY KEY, deptName VARCHAR(50) NOT NULL, mgrEmpNo VARCHAR(10), FOREIGN KEY (mgrEmpNo) REFERENCES employee(empNo));
```

Query OK, 0 rows affected (0.09 sec)

```
DESCRIBE department;
```

Field	Type	Null	Key	Default	Extra
deptNo	varchar(10)	NO	PRI	NULL	
deptName	varchar(50)	NO		NULL	
mgrEmpNo	varchar(10)	YES	MUL	NULL	

3 rows in set (0.00 sec)

```
CREATE TABLE project (projNo VARCHAR(10) NOT NULL PRIMARY KEY, projName VARCHAR(50) NOT NULL, deptNo VARCHAR(10) NOT NULL, FOREIGN KEY (deptNo) REFERENCES department(deptNo));
Query OK, 0 rows affected (0.06 sec)
```

```
DESCRIBE project;
```

Field	Type	Null	Key	Default	Extra
projNo	varchar(10)	NO	PRI	NULL	
projName	varchar(50)	NO		NULL	
deptNo	varchar(10)	NO	MUL	NULL	

3 rows in set (0.01 sec)

```
CREATE TABLE works_on (empNo VARCHAR(10) NOT NULL, projNo VARCHAR(10) NOT NULL, dateWorked DATE NOT NULL, hoursWorked INT, CONSTRAINT Chk_hoursWorked CHECK (hoursWorked BETWEEN 0 AND 40), PRIMARY KEY (empNo, projNo, dateWorked), FOREIGN KEY (empNo) REFERENCES employee(empNo), FOREIGN KEY (projNo) REFERENCES project(projNo));
Query OK, 0 rows affected (0.11 sec)
```

```
DESCRIBE works_on;
```

Field	Type	Null	Key	Default	Extra
empNo	varchar(10)	NO	PRI	NULL	
projNo	varchar(10)	NO	PRI	NULL	

dateWorked	date	NO	PRI	NULL
hoursWorked	int	YES		NULL

4 rows in set (0.01 sec)

```
INSERT INTO employee (empNo, fName, lName, address, DOB, sex, position, deptNo)
VALUES
('EMP001','Kwame','Mensah','Accra, Ghana','1985-03-12','M','Manager','DEP001'),
('EMP002','Ama','Owusu','Kumasi, Ghana','1990-07-21','F','Manager','DEP002'),
('EMP003','Kofi','Asare','Takoradi, Ghana','1992-11-05','M','Analyst','DEP002'),
('EMP004','Akosua','Boateng','Cape Coast, Ghana','1988-04-17','F','Manager','DEP001'),
('EMP005','Yaw','Appiah','Tema, Ghana','1994-01-30','M','Software Developer','DEP001'),
('EMP006','Adwoa','Frimpong','Sunyani, Ghana','1991-09-14','F','Manager','DEP006'),
('EMP007','Kojo','Antwi','Tamale, Ghana','1987-06-02','M','Team Leader','DEP007'),
('EMP008','Efua','Bonsu','Ho, Ghana','1993-12-09','F','Manager','DEP008'),
('EMP009','Kwabena','Darko','Koforidua, Ghana','1986-08-25','M','Manager','DEP004'),
('EMP010','Abena','Osei','Kasoa, Ghana','1995-02-18','F','Software Developer','DEP010');
```

Query OK, 10 rows affected (0.02 sec)

Records: 10 Duplicates: 0 Warnings: 0

SELECT * FROM employee;

empNo	fName	lName	address	DOB	sex	position	deptNo
EMP001	Kwame	Mensah	Accra, Ghana	1985-03-12	M	Manager	DEP001
EMP002	Ama	Owusu	Kumasi, Ghana	1990-07-21	F	Manager	DEP002
EMP003	Kofi	Asare	Takoradi, Ghana	1992-11-05	M	Analyst	DEP002
EMP004	Akosua	Boateng	Cape Coast, Ghana	1988-04-17	F	Manager	DEP001
EMP005	Yaw	Appiah	Tema, Ghana	1994-01-30	M	Software Developer	DEP001
EMP006	Adwoa	Frimpong	Sunyani, Ghana	1991-09-14	F	Manager	DEP006
EMP007	Kojo	Antwi	Tamale, Ghana	1987-06-02	M	Team Leader	DEP007
EMP008	Efua	Bonsu	Ho, Ghana	1993-12-09	F	Manager	DEP008
EMP009	Kwabena	Darko	Koforidua, Ghana	1986-08-25	M	Manager	DEP004
EMP010	Abena	Osei	Kasoa, Ghana	1995-02-18	F	Software Developer	DEP010

10 rows in set (0.00 sec)

```
INSERT INTO department (deptNo, deptName, mgrEmpNo)
VALUES
```

```
('DEP001','Software Development','EMP001'),
('DEP002','Data Analytics','EMP002'),
('DEP003','Human Resources','EMP003'),
('DEP004','Finance','EMP009'),
('DEP005','Operations','EMP005'),
('DEP006','Research','EMP006'),
('DEP007','Cyber Security','EMP007'),
('DEP008','Marketing','EMP008'),
('DEP009','Customer Support','EMP004'),
('DEP010','Quality Assurance','EMP010');
```

Query OK, 10 rows affected (0.04 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
SELECT * FROM department;
```

deptNo	deptName	mgrEmpNo
DEP001	Software Development	EMP001
DEP002	Data Analytics	EMP002
DEP003	Human Resources	EMP003
DEP004	Finance	EMP009
DEP005	Operations	EMP005
DEP006	Research	EMP006
DEP007	Cyber Security	EMP007
DEP008	Marketing	EMP008
DEP009	Customer Support	EMP004
DEP010	Quality Assurance	EMP010

10 rows in set (0.00 sec)

```

INSERT INTO project (projNo, projName, deptNo)
VALUES
('PRJ001','Bienvenitu Platform','DEP001'),
('PRJ002','Student Portal Analytics','DEP002'),
('PRJ003','Recruitment System','DEP003'),
('PRJ004','Payroll Automation','DEP004'),
('PRJ005','Inventory Tracker','DEP005'),
('PRJ006','AI Research Hub','DEP006'),
('PRJ007','Security Monitoring Tool','DEP007'),
('PRJ008','Digital Marketing Campaign','DEP008'),
('PRJ009','Customer Feedback System','DEP009'),
('PRJ010','Software Testing Suite','DEP010');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

```

```

SELECT * FROM project;

```

projNo	projName	deptNo
PRJ001	Bienvenitu Platform	DEP001
PRJ002	Student Portal Analytics	DEP002
PRJ003	Recruitment System	DEP003
PRJ004	Payroll Automation	DEP004
PRJ005	Inventory Tracker	DEP005
PRJ006	AI Research Hub	DEP006
PRJ007	Security Monitoring Tool	DEP007
PRJ008	Digital Marketing Campaign	DEP008
PRJ009	Customer Feedback System	DEP009
PRJ010	Software Testing Suite	DEP010

10 rows in set (0.00 sec)

```

INSERT INTO works_on (empNo, projNo, dateWorked, hoursWorked)
VALUES
('EMP001','PRJ001','2026-06-01',35),

```

```

('EMP002','PRJ002','2026-06-02',30),
('EMP003','PRJ002','2026-06-03',28),
('EMP004','PRJ001','2026-06-04',40),
('EMP005','PRJ005','2026-06-05',25),
('EMP006','PRJ006','2026-06-06',32),
('EMP007','PRJ007','2026-06-07',38),
('EMP008','PRJ008','2026-06-08',27),
('EMP009','PRJ004','2026-06-09',34),
('EMP010','PRJ010','2026-06-10',36);
Query OK, 10 rows affected (0.04 sec)
Records: 10  Duplicates: 0  Warnings: 0

```

```
SELECT * FROM works_on;
```

empNo	projNo	dateWorked	hoursWorked
EMP001	PRJ001	2026-06-01	35
EMP002	PRJ002	2026-06-02	30
EMP003	PRJ002	2026-06-03	28
EMP004	PRJ001	2026-06-04	40
EMP005	PRJ005	2026-06-05	25
EMP006	PRJ006	2026-06-06	32
EMP007	PRJ007	2026-06-07	38
EMP008	PRJ008	2026-06-08	27
EMP009	PRJ004	2026-06-09	34
EMP010	PRJ010	2026-06-10	36

```
10 rows in set (0.00 sec)
```

7.25

```

CREATE VIEW female_manager_projects AS
SELECT
    p.projNo,
    p.projName,
    d.deptNo,

```

```

        d.deptName,
        e.empNo AS managerNo,
        e.fName,
        e.lName
FROM project p
JOIN department d
    ON p.deptNo = d.deptNo
JOIN employee e
    ON d.mgrEmpNo = e.empNo
WHERE e.sex = 'F'
AND e.position = 'Manager'
ORDER BY p.projNo;
Query OK, 0 rows affected (0.04 sec)

```

```
SELECT * FROM female_manager_projects;
```

projNo	projName	deptNo	deptName	managerNo	fName	lName
PRJ002	Student Portal Analytics	DEP002	Data Analytics	EMP002	Ama	Owusu
PRJ006	AI Research Hub	DEP006	Research	EMP006	Adwoa	Frimpong
PRJ008	Digital Marketing Campaign	DEP008	Marketing	EMP008	Efua	Bonsu
PRJ009	Customer Feedback System	DEP009	Customer Support	EMP004	Akosua	Boateng

```
4 rows in set (0.03 sec)
```

7.26

```

CREATE VIEW employee_projects AS
SELECT
    e.empNo,
    e.fName,
    e.lName,
    p.projName,
    w.hoursWorked
FROM employee AS e
JOIN works_on AS w

```

```

        ON e.empNo = w.empNo
JOIN project AS p
        ON p.projNo = w.projNo;
Query OK, 0 rows affected (0.02 sec)

```

```

SELECT * FROM employee_projects;

```

empNo	fName	lName	projName	hoursWorked
EMP001	Kwame	Mensah	Bienvenitu Platform	35
EMP002	Ama	Owusu	Student Portal Analytics	30
EMP003	Kofi	Asare	Student Portal Analytics	28
EMP004	Akosua	Boateng	Bienvenitu Platform	40
EMP005	Yaw	Appiah	Inventory Tracker	25
EMP006	Adwoa	Frimpong	AI Research Hub	32
EMP007	Kojo	Antwi	Security Monitoring Tool	38
EMP008	Efua	Bonsu	Digital Marketing Campaign	27
EMP009	Kwabena	Darko	Payroll Automation	34
EMP010	Abena	Osei	Software Testing Suite	36

```

10 rows in set (0.00 sec)

```

7.27

```

CREATE VIEW EmpProject(empNo, projNo, totalHours) AS

```

```

SELECT

```

```

        w.empNo,
        w.projNo,
        SUM(hoursWorked) AS `totalHours`

```

```

FROM

```

```

        employee e,
        project p,
        works_on w

```

```

WHERE

```

```

        e.empNo = w.empNo
        AND p.projNo = w.projNo

```


GROUP BY

w.empNo,
w.projNo;

Query OK, 0 rows affected (0.04 sec)

a)

SELECT * FROM EmpProject;

empNo	projNo	totalHours
EMP001	PRJ001	35
EMP002	PRJ002	30
EMP003	PRJ002	28
EMP004	PRJ001	40
EMP005	PRJ005	25
EMP006	PRJ006	32
EMP007	PRJ007	38
EMP008	PRJ008	27
EMP009	PRJ004	34
EMP010	PRJ010	36

10 rows in set (0.03 sec)

b)

SELECT projNo FROM EmpProject WHERE projNo = 'SCCS';

Empty set (0.00 sec)

c)

SELECT COUNT(projNo) FROM EmpProject WHERE empNo = 'E1';

COUNT(projNo)
0

1 row in set (0.03 sec)

d)

```
SELECT empNo, totalHours FROM EmpProject GROUP BY empNo;
```

```
+-----+-----+
| empNo | totalHours |
+-----+-----+
| EMP001 |          35 |
| EMP002 |          30 |
| EMP003 |          28 |
| EMP004 |          40 |
| EMP005 |          25 |
| EMP006 |          32 |
| EMP007 |          38 |
| EMP008 |          27 |
| EMP009 |          34 |
| EMP010 |          36 |
+-----+-----+
10 rows in set (0.00 sec)
```

General

7.28

```
CREATE TABLE part (partNo VARCHAR(5) NOT NULL, contract VARCHAR(50) NOT NULL, partCost INT NOT NULL, PRIMARY KEY
(partNo, contract));
```

Query OK, 0 rows affected (0.08 sec)

```
INSERT INTO part (partNo, contract, partCost)
```

```
VALUES
```

```
('P001', 'Supply of Cement', 1200),
('P002', 'Electrical Installation', 500),
('P003', 'Road Construction', 12500),
('P004', 'Plumbing Services', 800),
('P005', 'Building Renovation', 9350),
('P006', 'Furniture Supply', 600),
('P007', 'Office Equipment', 6400),
('P008', 'Network Installation', 2500),
```

```

('P009', 'Painting Services', 3200),
('P010', 'Air Conditioning Installation', 700);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

```

```
SELECT * FROM part;
```

partNo	contract	partCost
P001	Supply of Cement	1200
P002	Electrical Installation	500
P003	Road Construction	12500
P004	Plumbing Services	800
P005	Building Renovation	9350
P006	Furniture Supply	600
P007	Office Equipment	6400
P008	Network Installation	2500
P009	Painting Services	3200
P010	Air Conditioning Installation	700

```
10 rows in set (0.00 sec)
```

```
CREATE VIEW ExpensiveParts (partNo) AS SELECT DISTINCT partNo FROM part WHERE partCost > 1000;
Query OK, 0 rows affected (0.03 sec)
```

```
SELECT * FROM ExpensiveParts;
```

partNo
P001
P003
P005
P007
P008
P009

6 rows in set (0.04 sec)

A materialized view physically stores the results of a query. The ExpensiveParts materialized view would store all part numbers with a partCost greater than 1000.

The view is maintained by updating it whenever the Part table changes:

Add a part when a new record with partCost > 1000 is inserted.

Add or remove a part when an update causes its cost to cross the 1000 threshold.

Remove a part when a qualifying record is deleted.

The materialized view can be maintained without accessing the underlying Part table if the database uses incremental maintenance, where the inserted, updated, or deleted row values are available through transaction logs or triggers. Only the affected rows are processed, avoiding a full scan of the base table.